# Increasing Array

You are given an array of nnn integers. You want to modify the array so that it is increasing, i.e., every element is at least as large as the previous element.

On each move, you may increase the value of any element by one. What is the minimum number of moves required?(print minimum required moves)

# Input

The first input line contains an integer n: the size of the array.

Then, the second line contains nnn integers x1,x2,…,xn: the contents of the array.

## example

Input: 5

3 2 5 1 7

Output:  5

## Approach

The problem can be solved by traversing the array and comparing each element with its previous element.
If the current element is smaller than the previous element, then it must be increased until it becomes equal to the previous element.
This requires (previous element - current element) increment operations.
By summing up the required operations for all such cases, we obtain the minimum number of moves needed to make the entire array non-decreasing.